
Cultural tendencies in generative AI

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Cultural Tendencies in Generative AI

Supplementary Information

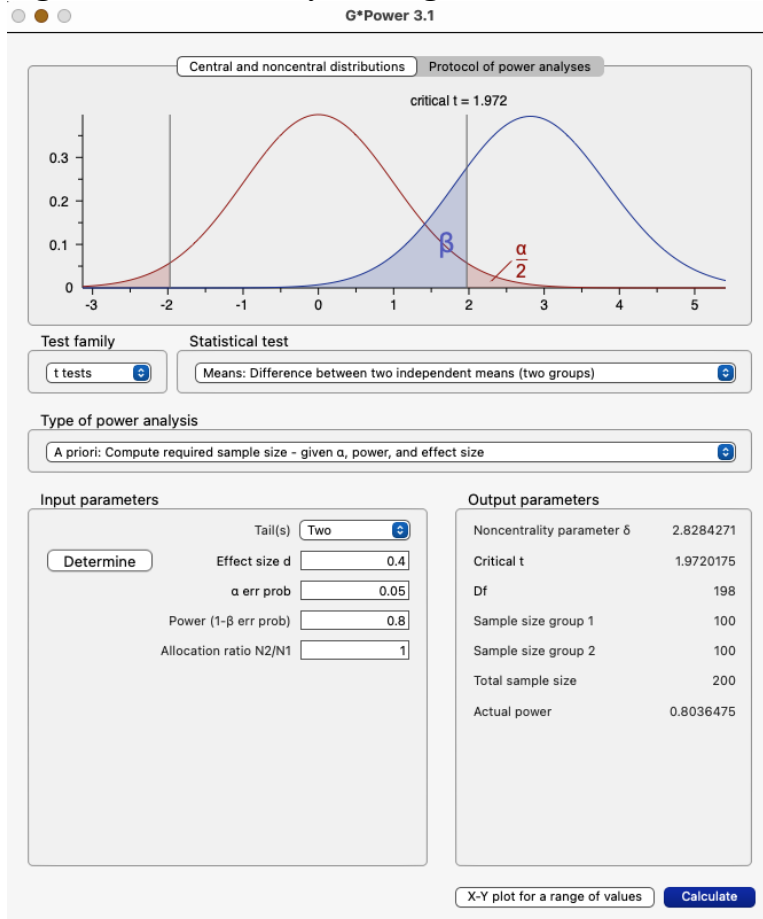
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SI Sec. 1. Power Analysis Using G*Power to Determine Sample Size

We used the G*Power 3.1 software to determine the sample size needed for a small-sized effect ($d = 0.4$): 100 iterations per language were needed for two-sided independent-samples t tests ($df = 198$) to be powered at 80% (see Figure S1 for power analysis). Thus, for each measure, we ran 100 iterations for the English version and 100 iterations for the Chinese version (i.e., $N = 200$).

Figure S1. Power analysis using G*Power 3.1



SI Sec. 2. Robustness Checks for Main Analyses

For the purpose of robustness, we conducted four additional sets of analyses: (1) forcing GPT to respond with only one numeric score, (2) setting GPT's temperature parameter to 1, (3) changing gender pronouns in vignette tasks, and (4) varying prompt formats (e.g., replacing *space* with *tab*; replacing *colon* with *dash*).

SI Sec. 2.1. Forcing GPT to respond with only one numeric score (Table S1)

Table S1. As a robustness check, we conducted another full set of analyses with identical prompts, except that **we forced GPT to respond with a single numeric score**. When used in Chinese (vs. English), GPT overall exhibited a more interdependent (vs. independent) social orientation and a more holistic (vs. analytic) cognitive style.

	Measure	Mean (SD)		Significance test
		Chinese	English	
Social orientation (interdependent vs. independent orientation)	Collectivism Scale (Yamaguchi, 1994)	4.83 (1.79)	4.07 (1.14)	$t = 11.35, p < .001, d = .51$
	Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	5.17 (.94)	3.83 (.37)	$t = 30.70, p < .001, d = 1.87$
	Individual–Collective Primacy Scale (Chen et al., 1998)	4.98 (.38)	4.37 (.61)	$t = 18.76, p < .001, d = 1.19$
	Inclusion of Other in the Self Scale (Aron et al., 1992; Li et al., 2006)	3.27 (.13)	2.38 (.52)	$t = 16.85, p < .001, d = 2.38$
Cognitive style (holistic vs. analytic)	Attribution bias task (Kitayama et al., 2006)	.18 (1.86)	-.22 (2.12)	$t = 4.88, p < .001, d = .20$
	Intuitive (vs. formal) reasoning task (Norenzayan et al., 2002)	.74 (.04)	.50 (.00)	Poisson regression: $B = .40, SE = .18, z = 2.18, p = .029$
	Expectation of change task (Ji et al., 2001)	.50 (.00)	.33 (.01)	beta regression: $B = .70, SE = .00, z = 178.03, p < .001$

SI Sec. 2.2. Setting GPT's temperature parameter to 1 (Tables S2-S4)

Table S2. As a robustness check, we repeated all analyses with **temperature set to 1** to increase response variability in GPT's output. When used in Chinese (vs. English), GPT overall exhibited a more interdependent (vs. independent) social orientation and a more holistic (vs. analytic) cognitive style.

	Measure	Mean (SD)		Significance test
		Chinese	English	
Social orientation (interdependent vs. independent orientation)	Collectivism Scale (Yamaguchi, 1994)	4.98 (.48)	4.08 (.77)	$t = 8.04, p < .001, d = 1.40$
	Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	4.39 (.64)	4.08 (.39)	$t = 3.34, p = .001, d = .58$
	Individual–Collective Primacy Scale (Chen et al., 1998)	4.88 (.44)	4.69 (.47)	$t = 1.82, p = .077, d = .41$
	Inclusion of Other in the Self Scale (Aron et al., 1992; Li et al., 2006)	3.57 (.75)	3.24 (.54)	$t = 3.54, p < .001, d = .50$
Cognitive style (holistic vs. analytic)	Attribution bias task (Kitayama et al., 2006)	-1.28 (1.82)	-1.97 (1.60)	$t = 9.78, p < .001, d = .40$
	Intuitive (vs. formal) reasoning task (Norenzayan et al., 2002)	3.05 (.74)	1.54 (.95)	Poisson regression: $B = .68, SE = .10, z = 6.91, p < .001$
	Expectation of change task (Ji et al., 2001)	.36 (.05)	.28 (.04)	beta regression: $B = .40, SE = .03, z = 13.97, p < .001$

Table S3. As a robustness check, we repeated all analyses with **temperature set to 1** to increase response variability in GPT’s output. When used in Chinese (vs. English), GPT overall exhibited a more holistic (vs. analytic) cognitive style: GPT was more likely to provide context-sensitive answers and range scores (vs. single score).

Measures	Context-Sensitive Answers				Range Scores (vs. Single Score)			
	Chinese	English	χ^2	<i>p</i> value	Chinese	English	χ^2	<i>p</i> value
Collectivism Scale (Yamaguchi, 1994)	25%	0%	28.57	< .001	28%	0%	32.56	< .001
Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	5%	0%	5.13	.024	14%	0%	15.05	< .001
Individual–Collective Primacy Scale (Chen et al., 1998)	48%	4%	50.31	< .001	41%	3%	42.07	< .001
Inclusion of Other in the Self Scale (Aron et al., 1992; Li et al., 2006)	27%	13%	6.12	.013	66%	44%	9.78	.002

Table S4. As a robustness check, we set GPT's temperature parameter to 1 and forced GPT to respond with a single numeric score. When used in Chinese (vs. English), GPT overall exhibited a more interdependent (vs. independent) social orientation and a more holistic (vs. analytic) cognitive style.

	Measure	Mean (SD)		Significance test
		Chinese	English	
Social orientation (independent vs. interdependent orientation)	Collectivism Scale (Yamaguchi, 1994)	4.84 (1.69)	4.10 (1.18)	$t = 11.23, p < .001, d = .51$
	Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	4.90 (1.04)	3.91 (.48)	$t = 20.58, p < .001, d = 1.23$
	Individual–Collective Primacy Scale (Chen et al., 1998)	4.80 (.64)	4.38 (.66)	$t = 10.14, p < .001, d = .65$
	Inclusion of Other in the Self Scale (Aron et al., 1992; Li et al., 2006)	3.65 (.52)	2.67 (.56)	$t = 12.80, p < .001, d = 1.82$
Cognitive style (analytic vs. holistic)	Attribution bias task (Kitayama et al., 2006)	.10 (1.92)	-.13 (2.05)	$t = 2.92, p = .004, d = .12$
	Intuitive (vs. formal) reasoning task (Norenzayan et al., 2002)	.72 (.08)	.49 (.07)	Poisson regression: $B = .39, SE = .19, z = 2.08, p = .037$
	Expectation of change task (Ji et al., 2001)	.50 (.01)	.33 (.02)	beta regression: $B = .69, SE = .01, z = 62.63, p < .001$

SI Sec. 2.3. Changing gender pronouns in vignette tasks

Table S5. As a robustness check, we changed **gender pronouns in vignette tasks**. When used in Chinese (vs. English), GPT overall exhibited a more holistic (vs. analytic) cognitive style: GPT was more likely to (a) engage in situational (vs. dispositional) attribution and (b) predict changes in the future.

	Mean (SD)		Significance test
	Chinese	English	
Attribution bias task (Kitayama et al., 2006)	-1.54 (1.78)	-2.20 (1.42)	$t = 9.91, p < .001, d = .40$
Expectation of change task (Ji et al., 2001)	.37 (.06)	.31 (.07)	beta regression: $B = .25, SE = .04, z = 6.41, p < .001$

SI Sec. 2.4. Varying prompt formats (Tables S6-S12)

We also conducted a series of robustness checks by varying prompt formats (Sclar et al., 2024).

Consider the following prompt:

“Based on general conditions, please explicitly give only one score for each option: 1 = strongly disagree and 7 = strongly agree.

1. Individuals should sacrifice self-interest for the group.
2. Individuals should stick with the group even through difficulties.
3. Group welfare is more important than individual rewards.
4. Group success is more important than individual success.
5. Individuals should only pursue their goals after considering the welfare of the group.
6. Group loyalty should be encouraged even if individual goals suffer.”

(The order of the items was randomized for each iteration).

In programming language, this was:

“Based on general conditions, please explicitly give only one score for each option: 1 = strongly disagree and 7 = strongly agree.\n1. Individuals should sacrifice self-interest for the group.\n2. Individuals should stick with the group even through difficulties.\n3. Group welfare is more important than individual rewards.\n4. Group success is more important than individual success.\n5. Individuals should only pursue their goals after considering the welfare of the group.\n6. Group loyalty should be encouraged even if individual goals suffer.”

where “\n” represents a new line.

Based on Sclar et al. (2024), there are six types of “**prompt format space**” in this example:

“Based on general conditions, please explicitly give only one score for each option {S1} 1 = strongly disagree and 7 = strongly agree {C1} {F_{item}} {S2} Individuals should sacrifice self-interest for the group {C2} {F_{item}} {S2} Individuals should stick with the group even through difficulties {C2} {F_{item}} {S2} Group welfare is more important than individual rewards {C2} {F_{item}} {S2} Group success is more important than individual success {C2} {F_{item}} {S2} Individuals should only pursue their goals after considering the welfare of the group {C2} {F_{item}} {S2} Group loyalty should be encouraged even if individual goals suffer {C3}”

We conducted a series of robustness checks using different prompt formats, and results are substantively similar (Tables S6-S12).

Table S6. We conducted a series of robustness checks by varying **prompt formats**. The corresponding results are displayed in Tables S7-S12.

Prompt format space	Meaning	Prompt format in our main analyses	New prompt format 1 (robustness check)	New prompt format 2 (robustness check)	New prompt format 3 (robustness check)
{S1}	the separator between two fields of instruction	.	no space	,	--
{S2}	the separator between item number and item content	single space	no space	\t	double space
{C1}	the space between instruction and items	.\n	\n	...	...
{C2}	the space between items	.\n	\n	\n\t	:
{C3}	the space after the last item	.	no space	-	::
{F _{item} }	the format of each item number	1.	(1)	<1>	1)

Note. “\n” denotes new line. “\t” denotes tab. The various prompt formats were sourced from Sclar et al. (2024).

Table S7. As a robustness check, we used new prompt format 1 in Table S6. When used in Chinese (vs. English), GPT overall exhibited a more interdependent (vs. independent) social orientation and a more holistic (vs. analytic) cognitive style.

	Measure	Mean (SD)		Significance test
		Chinese	English	
Social orientation (independent vs. interdependent orientation)	Collectivism Scale (Yamaguchi, 1994)	4.72 (.56)	4.32 (.81)	$t = 3.11, p = .002, d = .58$
	Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	4.31 (.40)	4.15 (.35)	$t = 2.73, p = .007, d = .41$
	Individual–Collective Primacy Scale (Chen et al., 1998)	4.68 (.40)	4.40 (.46)	$t = 2.31, p = .031, d = .65$
	Inclusion of Other in the Self Scale (Aron et al., 1992; Li et al., 2006)	3.06 (.35)	2.81 (.14)	$t = 6.76, p < .001, d = .96$
Cognitive style (analytic vs. holistic)	Attribution bias task (Kitayama et al., 2006)	-1.41 (1.76)	-1.66 (1.39)	$t = 3.98, p < .001, d = .16$
	Intuitive (vs. formal) reasoning task (Norenzayan et al., 2002)	2.84 (.72)	1.88 (.88)	Poisson regression: $B = .41, SE = .09, z = 4.39, p < .001$
	Expectation of change task (Ji et al., 2001)	.37 (.06)	.30 (.05)	beta regression: $B = .30, SE = .04, z = 8.31, p < .001$

Table S8. As a robustness check, we used new prompt format 1 in Table S6. When used in Chinese (vs. English), GPT overall exhibited a more holistic (vs. analytic) cognitive style: GPT was more likely to provide context-sensitive answers and range scores (vs. single score).

Measures	Context-Sensitive Answers				Range Scores (vs. Single Score)			
	Chinese	English	χ^2	<i>p</i> value	Chinese	English	χ^2	<i>p</i> value
Collectivism Scale (Yamaguchi, 1994)	33%	0%	39.52	< .001	44%	1%	53.02	< .001
Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	2%	0%	2.02	.155	4%	0%	4.08	.043
Individual–Collective Primacy Scale (Chen et al., 1998)	81%	1%	132.29	< .001	54%	1%	70.45	< .001

Table S9. As a robustness check, we used new prompt format 2 in Table S6. When used in Chinese (vs. English), GPT overall exhibited a more interdependent (vs. independent) social orientation and a more holistic (vs. analytic) cognitive style.

	Measure	Mean (SD)		Significance test
		Chinese	English	
Social orientation (independent vs. interdependent orientation)	Collectivism Scale (Yamaguchi, 1994)	4.79 (.26)	4.21 (.82)	$t = 4.51, p < .001, d = .96$
	Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	4.33 (.39)	4.03 (.34)	$t = 5.33, p < .001, d = .80$
	Individual–Collective Primacy Scale (Chen et al., 1998)	4.69 (.36)	4.50 (.54)	$t = 1.81, p = .076, d = .41$
	Inclusion of Other in the Self Scale (Aron et al., 1992; Li et al., 2006)	3.26 (.07)	2.75 (.04)	$t = 67.15, p < .001, d = 9.50$
Cognitive style (analytic vs. holistic)	Attribution bias task (Kitayama et al., 2006)	-1.64 (1.54)	-1.66 (1.46)	$t = .37, p = .709, d = .02$
	Intuitive (vs. formal) reasoning task (Norenzayan et al., 2002)	2.27 (.89)	1.61 (.87)	Poisson regression: $B = .34, SE = .10, z = 3.33, p < .001$
	Expectation of change task (Ji et al., 2001)	.39 (.06)	.27 (.04)	beta regression: $B = .54, SE = .03, z = 16.33, p < .001$

Table S10. As a robustness check, we used new prompt format 2 in Table S6. When used in Chinese (vs. English), GPT overall exhibited a more holistic (vs. analytic) cognitive style: GPT was more likely to provide context-sensitive answers and range scores (vs. single score).

Measures	Context-Sensitive Answers				Range Scores (vs. Single Score)			
	Chinese	English	χ^2	<i>p</i> value	Chinese	English	χ^2	<i>p</i> value
Collectivism Scale (Yamaguchi, 1994)	38%	0%	46.91	< .001	37%	2%	39.02	< .001
Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	5%	0%	5.13	.024	8%	0%	8.33	.004
Individual–Collective Primacy Scale (Chen et al., 1998)	78%	16%	77.16	< .001	58%	14%	42.01	< .001

Table S11. As a robustness check, we used new prompt format 3 in Table S6. When used in Chinese (vs. English), GPT overall exhibited a more interdependent (vs. independent) social orientation and a more holistic (vs. analytic) cognitive style.

	Measure	Mean (SD)		Significance test
		Chinese	English	
Social orientation (independent vs. interdependent orientation)	Collectivism Scale (Yamaguchi, 1994)	4.78 (.24)	4.26 (.66)	$t = 4.74, p < .001, d = 1.03$
	Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	4.39 (.65)	3.94 (.30)	$t = 6.24, p < .001, d = .88$
	Individual–Collective Primacy Scale (Chen et al., 1998)	4.76 (.31)	4.61 (.32)	$t = 2.65, p = .009, d = .49$
	Inclusion of Other in the Self Scale (Aron et al., 1992; Li et al., 2006)	3.27 (.26)	3.20 (.15)	$t = 2.27, p = .025, d = .32$
Cognitive style (analytic vs. holistic)	Attribution bias task (Kitayama et al., 2006)	-1.49 (1.71)	-1.74 (1.51)	$t = 3.79, p < .001, d = .15$
	Intuitive (vs. formal) reasoning task (Norenzayan et al., 2002)	2.69 (.73)	1.88 (.94)	Poisson regression: $B = .36, SE = .10, z = 3.77, p < .001$
	Expectation of change task (Ji et al., 2001)	.36 (.05)	.29 (.07)	beta regression: $B = .33, SE = .04, z = 8.16, p < .001$

Table S12. As a robustness check, we **used new prompt format 3 in Table S6**. When used in Chinese (vs. English), GPT overall exhibited a more holistic (vs. analytic) cognitive style: GPT was more likely to provide context-sensitive answers and range scores (vs. single score).

Measures	Context-Sensitive Answers				Range Scores (vs. Single Score)			
	Chinese	English	χ^2	<i>p</i> value	Chinese	English	χ^2	<i>p</i> value
Collectivism Scale (Yamaguchi, 1994)	23%	0%	25.99	< .001	30%	0%	35.29	< .001
Individual–Collective Primacy Scale (Chen et al., 1998)	58%	9%	53.89	< .001	47%	9%	35.81	< .001

SI Sec. 3. The Real-World Impact of Generative AI's Cultural Tendencies

We asked GPT to advise a start-up on selecting an advertising appeal from two alternatives: one slogan has an independent social orientation that emphasizes personal benefits (e.g., “Your future, your peace of mind. Our insurance.”), whereas the other slogan has an interdependent social orientation that emphasizes collective benefits (e.g., “Your family’s future, your promise. Our insurance.”). To be robust, we utilized three pairs of advertising appeals for different products: (1) insurance, (2) shoes, and (3) toothbrush. Results show that, for each pair of advertising appeals, GPT was more likely to recommend the interdependent-oriented (vs. independent-oriented) appeal when used in Chinese (vs. English) (Table S13).

Table S13. For each pair of advertising appeals, GPT was more likely to recommend the interdependent-oriented (vs. independent-oriented) appeal when used in Chinese (vs. English).

Products	Chinese	English	χ^2	p value
Insurance	100%	0%	200.00	< .001
Shoes	80%	0%	133.33	< .001
Toothbrush	62%	0%	89.86	< .001

SI Sec. 4. Using Cultural Prompts to Adjust Cultural Tendencies in Generative AI

We explored a tendency adjustment strategy: cultural prompts. We conducted a new set of analyses in English, using identical prompts from the Main Analyses but adding a reference to the Chinese cultural context (i.e., “You are an average person born and living in China”) (Shanahan et al., 2023). Results show that, when we added this Chinese cultural prompt (vs. not), GPT’s responses in English exhibited a more interdependent social orientation and a more holistic cognitive style (Table S14).

Table S14. When we added the Chinese cultural prompt (vs. not), GPT’s responses in English overall exhibited a more interdependent social orientation and a more holistic cognitive style.

	Measure	Mean (SD)		Significance test
		With Chinese cultural prompt (in English)	Without any cultural prompts (in English)	
Social orientation (independent vs. interdependent orientation)	Collectivism Scale (Yamaguchi, 1994)	5.26 (.18)	4.38 (.80)	$t = 6.89, p < .001, d = 1.52$
	Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	5.09 (.26)	3.98 (.30)	$t = 27.59, p < .001, d = 3.91$
	Individual–Collective Primacy Scale (Chen et al., 1998)	5.50 (.22)	4.58 (.54)	$t = 14.17, p < .001, d = 2.23$
	Inclusion of Other in the Self Scale (Aron et al., 1992; Li et al., 2006)	3.07 (.21)	2.76 (.36)	$t = 7.33, p < .001, d = 1.04$
Cognitive style (analytic vs. holistic)	Attribution bias task (Kitayama et al., 2006)	-1.67 (1.57)	-2.11 (1.49)	$t = 7.03, p < .001, d = .29$
	Intuitive (vs. formal) reasoning task (Norenzayan et al., 2002)	1.70 (.76)	1.23 (.84)	Poisson regression: $B = .32, SE = .12, z = 2.73, p = .006$
	Expectation of change task (Ji et al., 2001)	.29 (.03)	.28 (.03)	beta regression: $B = .02, SE = .02, z = 1.10, p = .270$

SI Sec. 5. Hypothesized Cultural Tendencies Also Appear in ERNIE

The cultural tendencies also appeared in ERNIE, a popular generative AI model in China. Specifically, ERNIE also exhibited a more interdependent (vs. independent) social orientation and a more holistic (vs. analytic) cognitive style when used in Chinese than in English (Table S15).¹

Table S15. When used in Chinese (vs. English), **ERNIE** overall exhibited a more interdependent (vs. independent) social orientation and a more holistic (vs. analytic) cognitive style.

	Measure	Mean (SD)		Significance test
		Chinese	English	
Social orientation (interdependent vs. independent orientation)	Collectivism Scale (Yamaguchi, 1994)	4.55 (.37)	4.08 (.55)	$t = 7.23, p < .001, d = 1.02$
	Individual Cultural Values: Collectivism Scale (Yoo et al., 2011)	6.33 (.42)	5.76 (.49)	$t = 8.77, p < .001, d = 1.24$
	Individual–Collective Primacy Scale (Chen et al., 1998)	4.77 (.61)	3.90 (.71)	$t = 9.30, p < .001, d = 1.32$
Cognitive style (holistic vs. analytic)	Attribution bias task (Kitayama et al., 2006)	.44 (3.25)	-.69 (2.79)	$t = 9.17, p < .001, d = .37$
	Intuitive (vs. formal) reasoning task (Norenzayan et al., 2002)	1.99 (.85)	1.98 (.57)	Poisson regression: $B = .01, SE = .10, z = 0.05, p = .96$
	Expectation of change task (Ji et al., 2001)	.34 (.05)	.25 (.03)	beta regression: $B = .44, SE = .03, z = 14.16, p < .001$

¹ We did not use the Inclusion of Other in the Self Scale for ERNIE because it was unable to process images.